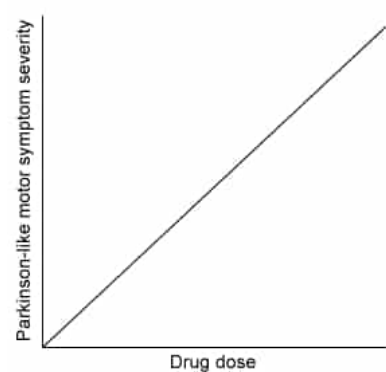


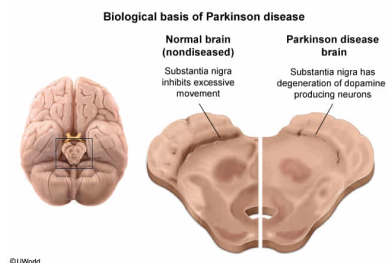
The graph below represents the relationship between the dose of a new drug and the severity of Parkinson-like motor symptoms, a suspected side effect. This drug is most likely a:



- ✓ ☒ A. dopamine antagonist. [64%]  
☐ B. dopamine agonist. [27%]  
☐ C. serotonin antagonist. [4%]  
☐ D. serotonin agonist. [3%]

Correct  
64% answered correctly

Explanation:



**Dopamine** is a neurotransmitter with widespread targets throughout the brain, specifically in the basal ganglia (**motor function**), **mesolimbic** pathway (pleasure, reward), and prefrontal cortex (motivation, emotion regulation).

**Parkinson disease** is associated with the **loss** of **dopaminergic neurons** in the **substantia nigra** (a structure in the basal ganglia that inhibits excess movement). A dopamine deficit in the basal ganglia causes motor symptoms such as resting tremors (eg, shaky hands), muscle rigidity, and shuffling gait.

Many antipsychotic drugs are dopamine **antagonists** that work in part by blocking the action of dopamine, an excess of which in certain areas of the brain has been linked to psychotic symptoms (eg, hallucinations), such as those seen in schizophrenia. These medications can have side effects that resemble the motor symptoms of Parkinson disease (eg, tremors, muscle rigidity).

A positive correlation between the dose of a drug and the severity of Parkinson-like motor symptoms (ie, severity of side effect symptoms increases as drug dose increases) suggests that this drug is most likely a **dopamine antagonist**, which blocks or reduces the effects of dopamine.

**(Choice B)** Some medications used to treat Parkinson disease are dopamine **agonists**, which mimic or enhance the effects of dopamine. These medications can include adverse side effects such as hallucinations or delusions (ie, psychotic symptoms).

**(Choices C and D)** Serotonin is a neurotransmitter involved in the experience of mood, sleep/wake regulation, social dominance/aggression, and appetite, among others. Neither a serotonin agonist nor antagonist would produce Parkinson-like motor symptoms.

Educational objective:

Dopamine is a neurotransmitter involved in movement, reward, pleasure, motivation, and emotion regulation. The loss of dopaminergic neurons in the substantia nigra associated with Parkinson disease causes motor symptoms such as resting tremors (eg, shaky hands), muscle rigidity, and shuffling gait.

Time spent: 0 sec  
Subject:  
Behavioral Sciences

QID: 400327  
System:  
Sensation, Perception, and Consciousness